

HARSH KAUSHIK

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Education

Dalhousie University

2026

Master of Computer Science

Halifax, Canada

- Conducted thesis research on image authenticity detection using complementary semantic and forensic representations to distinguish human-created, AI-generated, and AI-manipulated visual content.

GGSIPU University

2023

Bachelor of Technology, Electrical and Electronics Engineering (GPA: 3.48/4.0)

New Delhi

Skills

Languages: Python, C/C++, Java, JavaScript, SQL

Machine Learning & Computer Vision: Scikit-learn, NumPy, Pandas, TensorFlow, PyTorch, Pillow

Web & Backend: React, Node.js, Express.js, FastAPI, REST APIs

Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS (EC2, S3)

Experience

Nextria

Jan 2026 – Jul 2026

AI Research Intern

Halifax (remote)

- Developed **NexAI**, an image-authenticity analysis platform to distinguish human-created, AI-generated, and AI-manipulated visual content using complementary semantic and forensic evidence.
- Built scalable workflows for dataset preparation, controlled evaluation, model validation, and uncertainty-aware prediction across diverse image sources and transformation conditions.
- Implemented a real-time inference and application-integration framework producing interpretable, evidence-supported outputs; the work continues as product development and future research.

Dalhousie University

Jul 2022 – Sep 2022

Mitacs Research Intern

Halifax

- Conducted research at PINet Lab on machine learning-based security for IoT networks, including network traffic data collection and feature extraction.
- Developed and evaluated intrusion detection models for DoS/DDoS attacks, performing experimental analysis to improve robustness in real-world IoT environments.

Meraki Lab

Oct 2021 – Oct 2023

Product Engineer

New Delhi

- Designed and built a low-cost DIY EEG device for real-time brain signal acquisition, contributing to AI-based solutions for accessible neurotechnology and mental health applications.
- Developed deep reinforcement learning (DRL)-based neurofeedback systems for anxiety reduction, integrating signal processing with adaptive AI-driven feedback in a product-focused environment.

Publications

Evidence-Aware Image Authenticity Decision Framework

using SigLIP2 for Embeddings and Transparent Multi-Signal Fusion | *IEEE WIFS, 2026*

H. Kaushik, D. Upadhyay, Z. Hamel, R. Ramsaran, E. Kitchen, J. Manero, and S. Sampalli, "Evidence-Aware Image Authenticity Decision Framework using SigLIP2 for Embeddings and Transparent Multi-Signal Fusion," manuscript submitted to the IEEE International Workshop on Information Forensics and Security (WIFS), 2026.

Design and Development of a Solar-Based Wireless Electric Vehicle Charging System | *Springer, 2024*

S. Jain, S. Jain, S. Kumar, H. Kaushik, N. Nagpal, and R. Sharma, "Design and Development of a Solar-Based Wireless Electric Vehicle Charging System," in *Renewable Power for Sustainable Growth: Proceedings of the International Conference on Renewable Power (ICRP 2023)*, Lecture Notes in Electrical Engineering, vol. 1086, Springer, Singapore, pp. 481–494, 2024.

Hybrid Image Processing Device as Wearable Aide for Visually Impaired | *IEEE ICACCS, 2022*

A. Papanai and H. Kaushik, "Hybrid Image Processing Device as Wearable Aide for Visually Impaired," in *Proceedings of the 2022 8th International Conference on Advanced Computing and Communication Systems (ICACCS)*, 2022.

H. Kaushik, N. Sharma, and N. Pathak, “A Review of Technologies Trending Towards 5G Networks for Smart Healthcare Using IoT,” in *Futuristic Trends for Sustainable Development and Sustainable Ecosystems*, 2022.

Fellowships & Research Programs

NSERC CREATE Cybersecurity Program	Jun 2025 – Jul 2026
<i>Graduate Trainee</i>	<i>Dalhousie University</i>

- Member of the 2024–2026 NSERC CREATE Cybersecurity Program cohort, participating in cybersecurity research training, academic–industry collaboration, and networking with government, industry, and academic stakeholders.

Mitacs Globalink Graduate Fellowship Award	Jun 2023
<i>Fellowship Award</i>	<i>Mitacs Dalhousie University</i>

- Selected for the Mitacs Globalink Graduate Fellowship Award to continue graduate studies and research in Canada after completing the Mitacs Globalink Research Internship program.

Leadership Experience

Graduate Teaching Assistant and Marker	Sep 2024 – Apr 2026
<i>Faculty of Computer Science, Dalhousie University</i>	<i>Halifax, Nova Scotia</i>

- Served as a Graduate Teaching Assistant and Marker across three academic terms for Data Structures and Algorithms (CSCI 2110), Network Computing (CSCI 3171), and Advanced Network Security (CSCI 6708).
- Evaluated programming assignments and laboratory work, supported students with Java, data structures, algorithms, computer networks, and security concepts, and provided timely feedback using consistent assessment standards.